DWDM-LABEXPERIMENTS
Day3_exercise-1

Consider the data set and perform the Apriori Algorithm and FP algorithm support:3 and confidence=50%

Customer ID	Transaction ID	Items Bought
1	0001	$\{a, d, e\}$
1	0024	$\{a, b, c, e\}$
2	0012	$\{a, b, d, e\}$
2	0031	$\{a, c, d, e\}$
3	0015	$\{b, c, e\}$
3	0022	$\{b, d, e\}$
4	0029	$\{c, d\}$
4	0040	$\{a, b, c\}$
5	0033	$\{a, d, e\}$
5	0038	$\{a, b, e\}$

Consider the data set and perform the Apriori Algorithm and FP algorithm support:3 and confidence=50%

Aim:

To implement the Apriori Algorithm using the WEKA tool to identify frequent itemsets and generate association rules from a market basket dataset with minimum support of 3 and minimum confidence of 50%.

Procedure:

1. Open the WEKA application and select Explorer.
2. Click on the Preprocess tab and select Open file.
3. Load the downloaded apriori_fp_growth_dataset.arff dataset.
4. Verify that the dataset contains 10 transactions and 5 attributes.
5. Click on the Associate tab.
6. Observe the generated frequent itemsets and association rules.
7. Record the output and take a screenshot for documentation.

Figure 1: Loading the Market Basket Dataset in WEKA Explorer

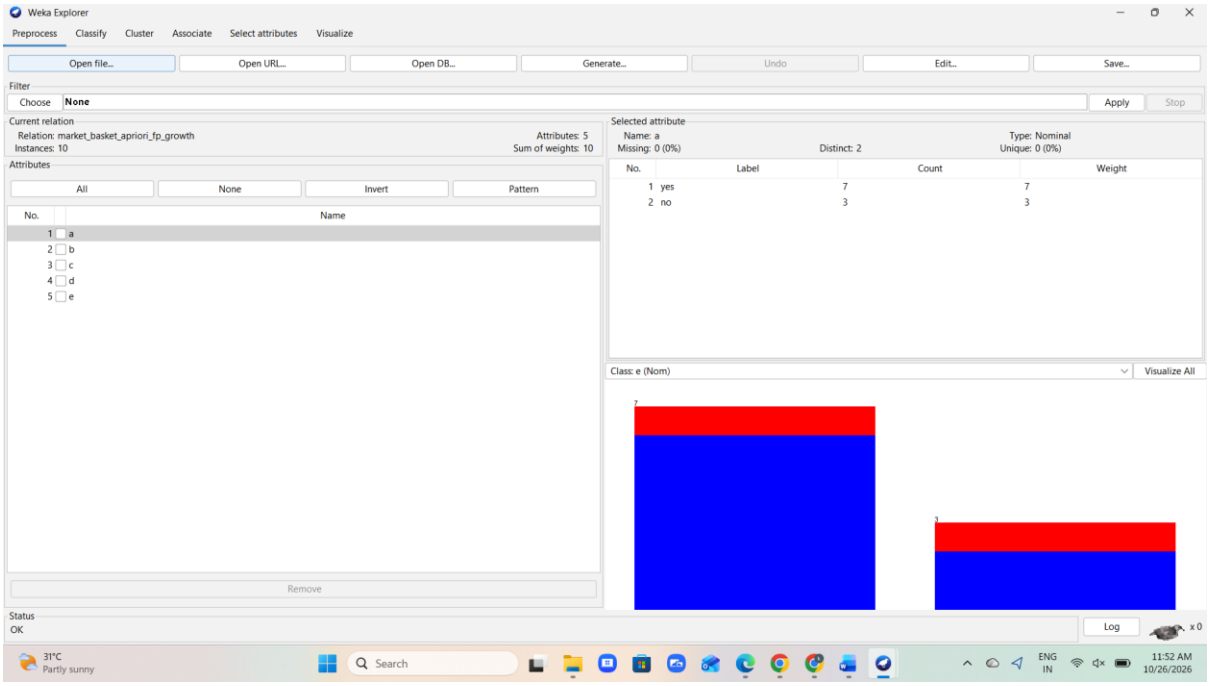


Figure 2: Execution of Apriori Algorithm in WEKA Explorer.

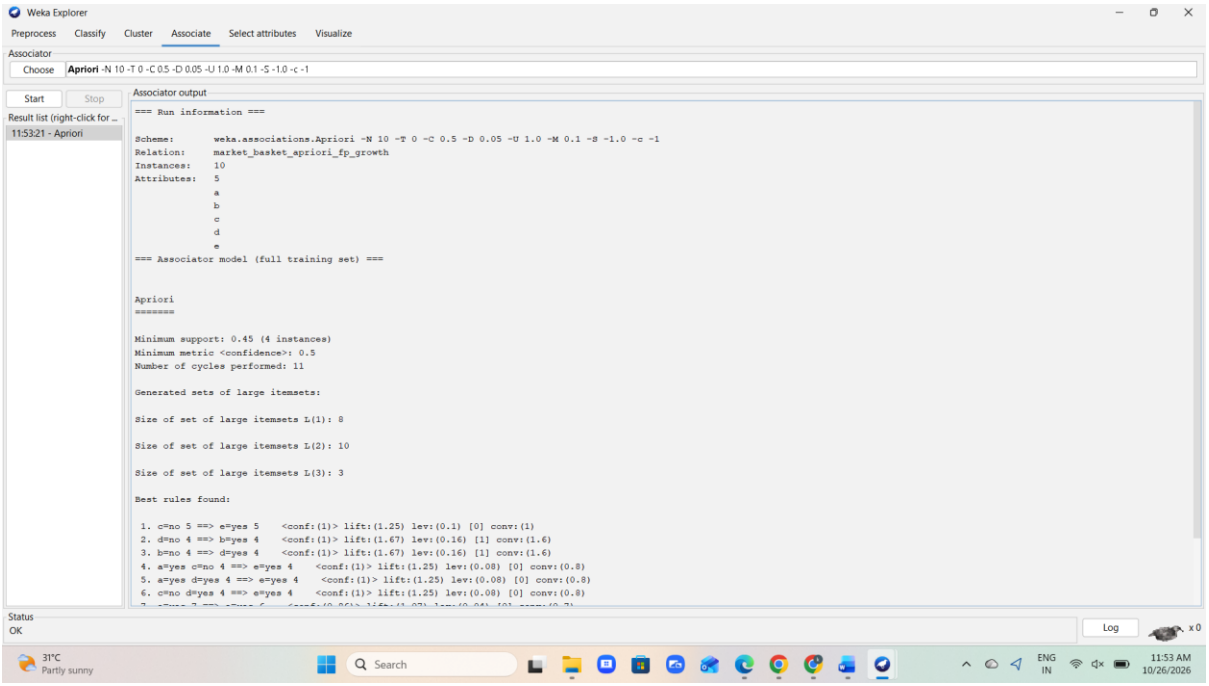


Figure 3: Output of Apriori Algorithm Showing Frequent Itemsets and Association Rules.

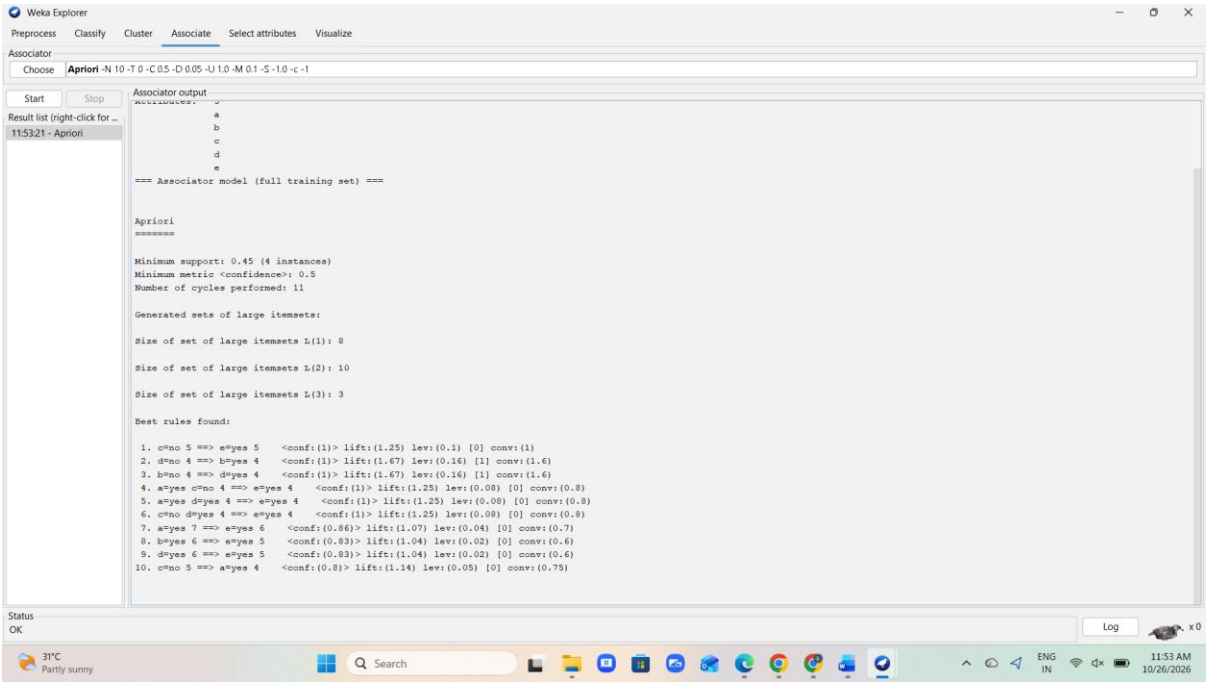
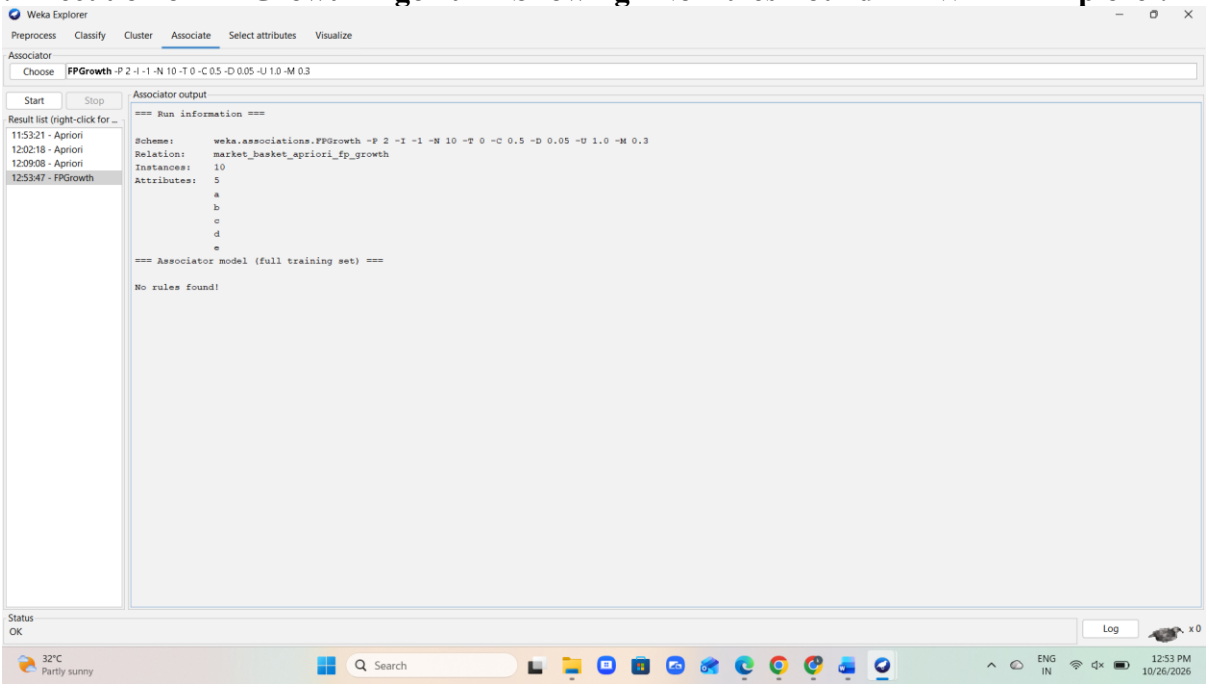


Figure 4
: Execution of FP-Growth Algorithm Showing "No Rules Found" in WEKA Explorer.



Result:

The dataset was successfully loaded into WEKA. The Apriori Algorithm was used to identify frequent itemsets and generate association rules based on the specified minimum support and confidence thresholds.

Consider the market basket transactions shown in the above table.

(a) What is the maximum number of association rules that can be extracted from this data (including rules that have zero support)?

(b) What is the maximum size of frequent itemsets that can be extracted (assuming minsup > 0)?

Transaction ID	Items Bought
1	{Milk, Beer, Diapers}
2	{Bread, Butter, Milk}
3	{Milk, Diapers, Cookies}
4	{Bread, Butter, Cookies}
5	{Beer, Cookies, Diapers}
6	{Milk, Diapers, Bread, Butter}
7	{Bread, Butter, Diapers}
8	{Beer, Diapers}
9	{Milk, Diapers, Bread, Butter}
10	{Beer, Cookies}

Aim:

To analyze a market basket dataset using WEKA and determine the maximum number of possible association rules and the maximum size of frequent itemsets.

Procedure

1. Open the WEKA application and select Explorer.
2. Click the Preprocess tab and select Open file.
3. Load the market_basket_six_items.arff dataset.
4. Verify that the dataset contains 10 instances and 6 attributes.
5. Observe the attribute names and their value distributions in the WEKA interface.
6. Determine the maximum number of association rules using the formula \(3^n-2^{n+1}+1\), where \(n\) is the number of distinct items.
7. Examine the transactions in the given dataset to identify the largest number of items appearing together.
8. Record the calculated results.

Figure 1: Loading the Market Basket Dataset in WEKA Explorer.

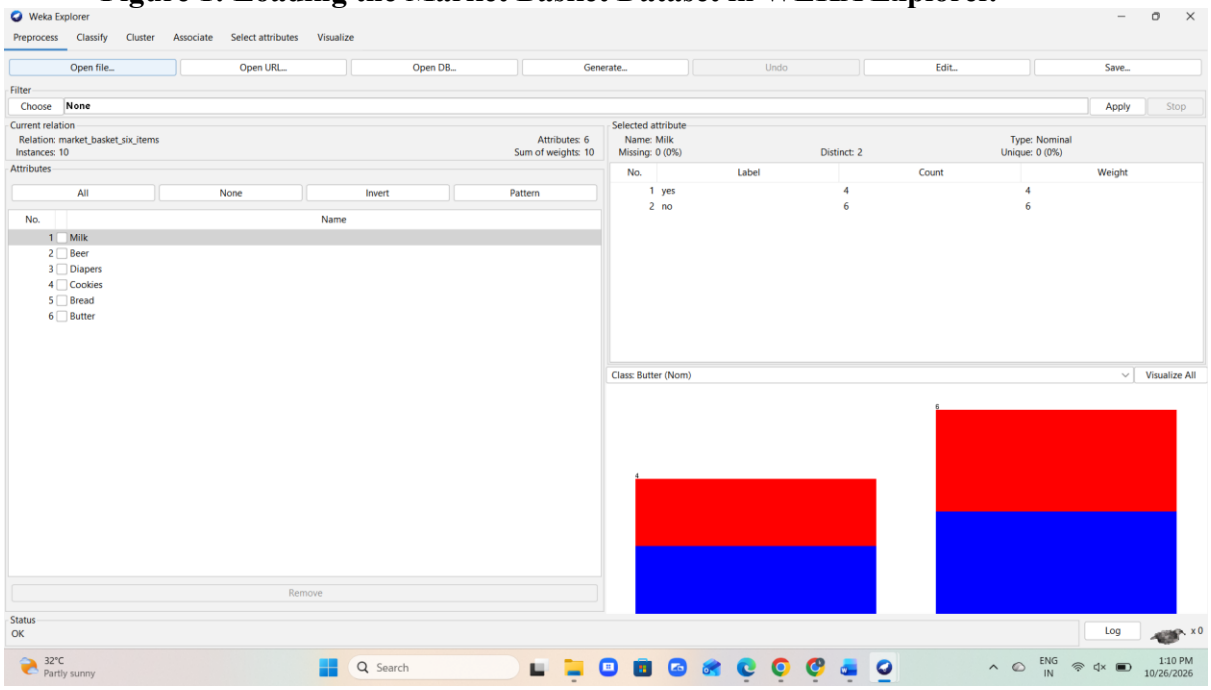


Figure 2: Apriori Algorithm Output Showing Frequent Itemsets and Association Rules in WEKA.

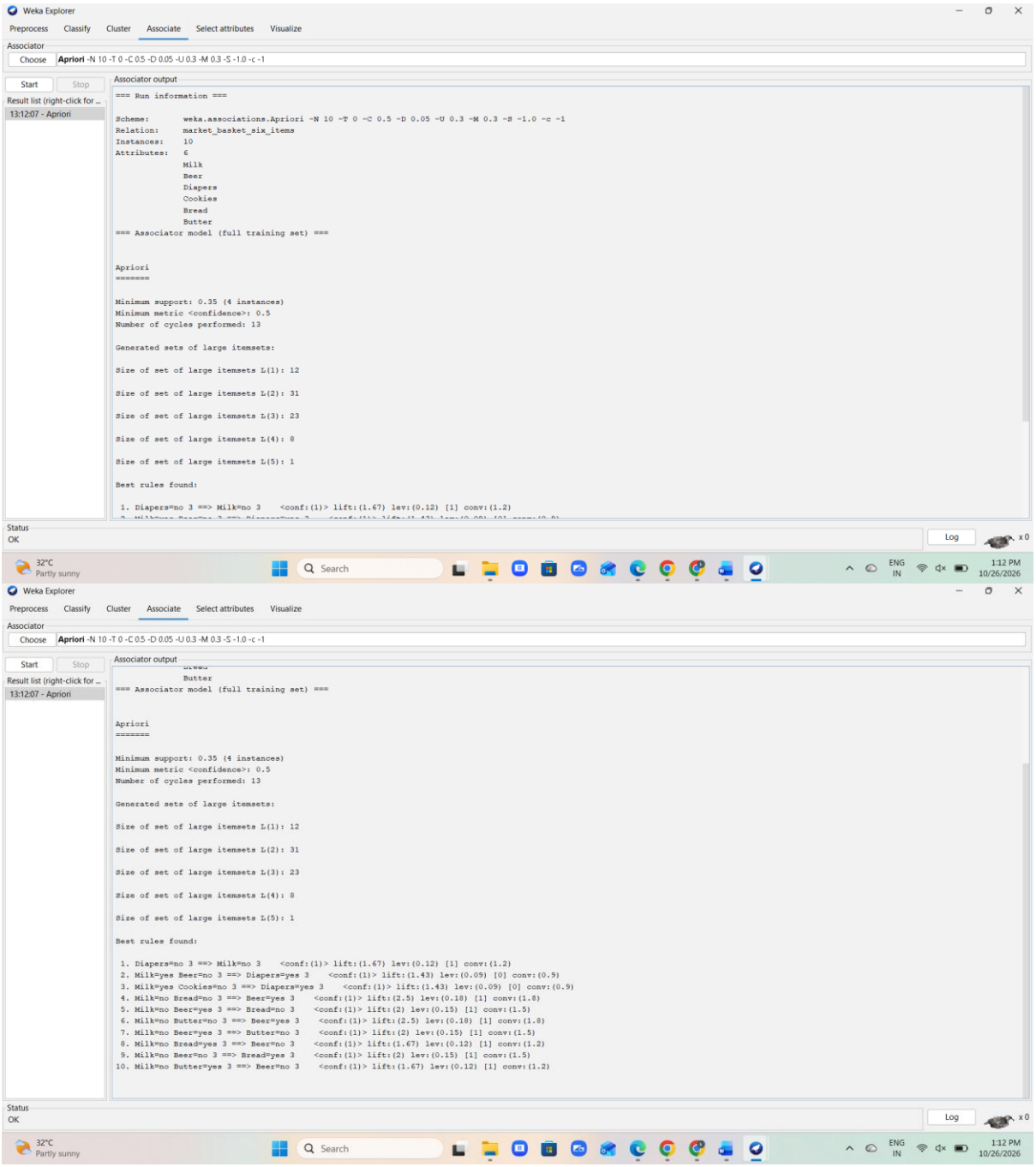
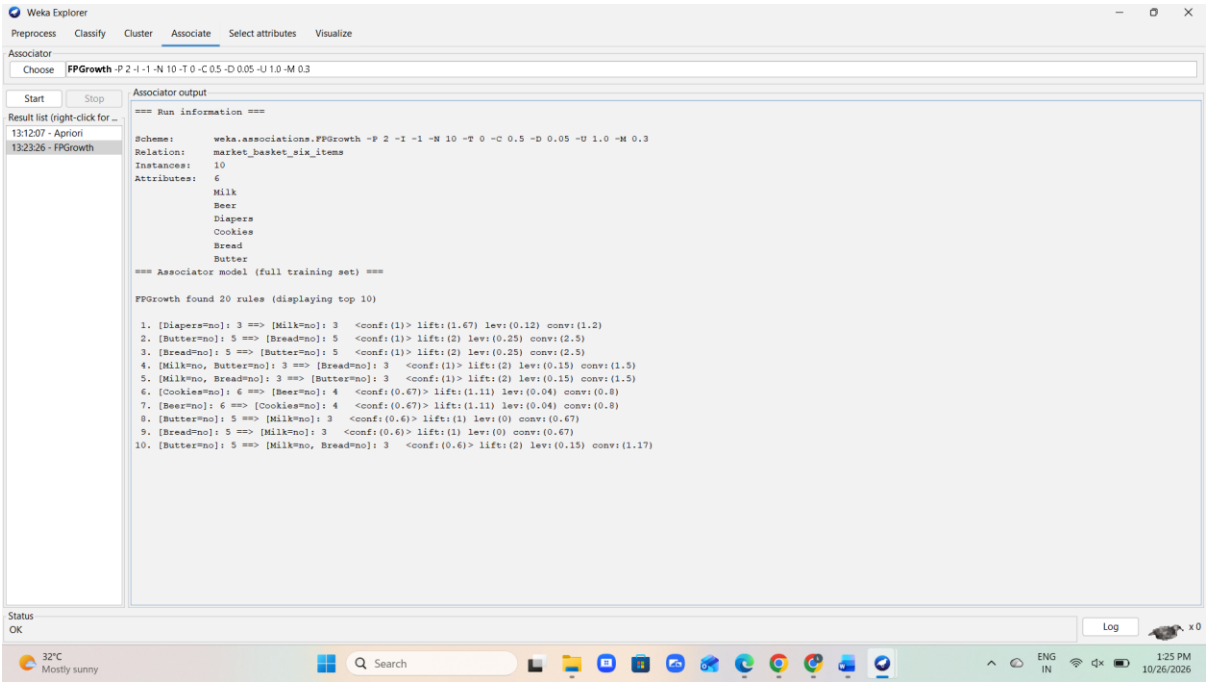


Figure 3: FP-Growth Algorithm Output Showing Association Rules for the Market Basket Dataset in WEKA Explorer.



Result

The market basket dataset was successfully loaded into WEKA, containing 10 transactions and 6 attributes. The maximum number of possible association rules for six distinct items is 602. For the original transaction table provided in the exercise, the maximum frequent itemset size is 4.